Altitudes, medians, and bisectors of triangles

ANSWER KEY

Core Concepts: Altitudes, medians, and bisectors of triangles

1 In $\triangle ABC$ studying altitudes, medians, and bisectors of triangles, $m\angle A = 42^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 93° ✓

b. 103°

c. 85°

d. 138°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (42^\circ + 45^\circ) = 93^\circ$.

2 A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm ✓

c. 18 cm

d. 24 cm

Multiplying the original length by the scale factor gives $6 \times 2 = 12$ cm.

3 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 1$?

a. 5

b. 10

c. 8 ✓

d. 11

Substituting $x = 1$ into $y = 3x + 5$ gives $y = 3(1) + 5 = 8$.

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Procedural Fluency & Algebraic Manipulation

4 In $\triangle ABC$ studying altitudes, medians, and bisectors of triangles, $m\angle A = 42^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 86° ✓

b. 96°

c. 78°

d. 138°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (42^\circ + 52^\circ) = 86^\circ$.

5 A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?

a. 15 cm

b. 18 cm ✓

c. 24 cm

d. 54 cm

Multiplying the original length by the scale factor gives $6 \times 3 = 18$ cm.

6 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 2$?

a. 8

b. 13

c. 11 ✓

d. 14

Substituting $x = 2$ into $y = 3x + 5$ gives $y = 3(2) + 5 = 11$.

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying altitudes, medians, and bisectors of triangles, $m\angle A = 42^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 79° ✓

b. 89°

c. 71°

d. 138°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (42^\circ + 59^\circ) = 79^\circ$.

8 A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length?

a. 20 cm

b. 24 cm ✓

c. 30 cm

d. 96 cm

Multiplying the original length by the scale factor gives $6 \times 4 = 24$ cm.

9 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 3$?

a. 11

b. 16

c. 14 ✓

d. 17

Substituting $x = 3$ into $y = 3x + 5$ gives $y = 3(3) + 5 = 14$.

Exam-Style Problem Solving & Modeling

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10 In $\triangle ABC$ studying altitudes, medians, and bisectors of triangles, $m\angle A = 42^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 72° ✓

b. 82°

c. 64°

d. 138°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (42^\circ + 66^\circ) = 72^\circ$.

11 A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length?

a. 25 cm

b. 30 cm ✓

c. 36 cm

d. 150 cm

Multiplying the original length by the scale factor gives $6 \times 5 = 30$ cm.

12 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 4$?

a. 14

b. 19

c. 17 ✓

d. 20

Substituting $x = 4$ into $y = 3x + 5$ gives $y = 3(4) + 5 = 17$.

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